

Polynomial and Hardy-equivalent cyclicity in $P^2(\mu)$

A candidate substantial partial answer to arXiv:1809.10182

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Status: candidate substantial partial result, likely valid. Under the full source hypotheses, every finite-codimensional shift-invariant subspace is cyclic, and these are exactly the invariant subspaces containing a nonzero polynomial. The full question is also affirmative for Hardy-equivalent measures, including every rotationally invariant example.

1. Source question and result

Let μ be a finite positive measure supported on $\overline{\mathbb{D}}$, let $P^2(\mu)$ be the closure of the analytic polynomials in $L^2(\mu)$, and let $Sf = zf$. Question 2.1 of Akeroyd–Conway–Yang [1] assumes that $P^2(\mu)$ is irreducible, $\text{abpe}(P^2(\mu)) = \mathbb{D}$, and $\mu(\mathbb{T}) > 0$, and asks whether every nonzero closed S -invariant subspace is cyclic.

$\mathbb{D} = \text{abpe}(P^2(\mu))$, $P^2(\mu)$ is irreducible and $\mu(\mathbb{T}) > 0$; and they did so with precision (see [4, Theorem 3.1]). This, by itself, is a very useful result. With (#) in hand, one can establish nontangential limits quite easily and the index one result is also within reach. Actually, Aleman, Richter and Sundberg establish the index one result using just the existence of nontangential limits (cf., [4, Theorem 3.2]). Therefore, Theorem III answers affirmatively Question 2.1 in [11], in its full generality. At the end of this section we give an alternate proof of Theorem III, part (b), that depends on (#). And, in Section 3, we provide an alternate route to the existence of nontangential limits that bypasses (#). All of these results seem to point in the direction of an affirmative answer to [11, Problem 2.2]. That is, if the hypothesis of Theorem III is in place and $\mathcal{M}$ is a nontrivial, closed invariant subspace for S_μ on $P^2(\mu)$, then there exists f in $P^2(\mu)$ such that $\mathcal{M} = [f]$ – the closure of $\{fp : p \text{ is a polynomial}\}$ in $L^2(\mu)$. In other words, the shift on every nontrivial, closed invariant subspace of $P^2(\mu)$ is cyclic. But very little progress has been made on this at all, not even in the case that $t = 2$. We pose this question once again and follow it with some observations.

Question 2.1. *Suppose that μ is supported in $\overline{\mathbb{D}}$, $P^2(\mu)$ is irreducible, $\text{abpe}(P^2(\mu)) = \mathbb{D}$ and $\mu(\mathbb{T}) > 0$. Is every nontrivial, closed, shift invariant subspace of $P^2(\mu)$ cyclic?*

Now, in a separate paper (cf., [3]), A. Aleman, S. Richter and C. Sundberg show

Figure 1: The exact cyclicity question on page 3 of arXiv:1809.10182.

For a polynomial q , let $q_{\mathbb{D}}$ be the monic polynomial obtained by retaining with multiplicity precisely the zeros of q in $\mathbb{D}$; it is 1 when there are none. Write $[f] = \overline{\{pf : p \in \mathbb{C}[z]\}}^{P^2(\mu)}$.

Theorem 1 (finite-codimensional classification). *Under the hypotheses above, for every nonzero polynomial q ,*

$$[q] = q_{\mathbb{D}} P^2(\mu), \quad \text{codim}[q] = \deg q_{\mathbb{D}}.$$

For a closed S -invariant subspace $M \neq \{0\}$, the following are equivalent:

1. M contains a nonzero polynomial;
2. M has finite codimension in $P^2(\mu)$;
3. $M = dP^2(\mu)$ for a monic polynomial d all of whose zeros lie in $\mathbb{D}$.

In particular every such M is cyclic, with generator d .

Theorem 2 (Hardy-equivalent measures). Write $\mu|_{\mathbb{T}} = h m$ and $\nu = \mu|_{\mathbb{D}}$. If

$$0 < c \leq h \leq C < \infty \quad (m\text{-a.e.})$$

and ν is an H^2 -Carleson measure, then every closed invariant subspace of $P^2(\mu)$ is cyclic. In particular this holds for every rotationally invariant μ satisfying the source hypotheses.

2. Boundary evaluations and division

The first point is that irreducibility removes a possible boundary obstruction.

Lemma 1 (no boundary bounded point evaluations). $\text{bpe}(P^2(\mu)) \cap \mathbb{T} = \emptyset$. Consequently $S - \zeta$ has dense range for every $\zeta \in \mathbb{T}$.

Proof. Fix $\zeta \in \mathbb{T}$ and set

$$u_n(z) = \left(\frac{1 + \bar{\zeta}z}{2} \right)^n.$$

On $\overline{\mathbb{D}}$, $|u_n| \leq 1$ and u_n tends pointwise to $\mathbf{1}_{\{\zeta\}}$. If $\mu(\{\zeta\}) = 0$, dominated convergence gives $\|u_n\|_{P^2(\mu)} \rightarrow 0$, whereas $u_n(\zeta) = 1$, ruling out bounded evaluation. If $\mu(\{\zeta\}) > 0$, then $u_n \rightarrow \mathbf{1}_{\{\zeta\}}$ in $L^2(\mu)$. This puts a nontrivial characteristic function in $P^2(\mu)$, contrary to irreducibility.

If $S - \zeta$ did not have dense range, some nonzero v would satisfy $S^*v = \bar{\zeta}v$. Then $\langle p, v \rangle = p(\zeta)\langle \mathbf{1}, v \rangle$ for polynomials p . Here $\langle \mathbf{1}, v \rangle \neq 0$, since otherwise density of the polynomials would give $v = 0$. Thus evaluation at ζ would be bounded, a contradiction. $\square$

Lemma 2 (bounded interior division). For $a \in \mathbb{D}$, the map

$$D_a p = \frac{p - p(a)}{z - a}$$

extends boundedly to $P^2(\mu)$. Hence

$$(S - a)P^2(\mu) = \ker(\text{ev}_a).$$

More generally, if all zeros of $r \in \mathbb{C}[z]$ lie in $\mathbb{D}$, then $rP^2(\mu)$ is closed, has codimension $\deg r$, and is described by vanishing of the corresponding zero jets.

Proof. Choose $0 < \rho < R$ with $\overline{B(a, R)} \subset \mathbb{D}$. Analytic bounded point evaluations are locally uniformly bounded, so

$$K = \sup_{|w-a|=R} \|\text{ev}_w\| < \infty.$$

The maximum principle applied to $D_a p$ gives, on $B(a, \rho)$,

$$|D_a p| \leq R^{-1}(K + \|\text{ev}_a\|)\|p\|_{P^2(\mu)}.$$

On its complement, $|z - a| \geq \rho$ and therefore

$$|D_a p(z)| \leq \rho^{-1}(|p(z)| + |p(a)|).$$

Integrating these two estimates proves $\|D_a p\|_{P^2(\mu)} \leq C_a \|p\|_{P^2(\mu)}$. Completion and locally uniform convergence identify the extension with analytic divided difference. Thus every f with $f(a) = 0$ equals $(z - a)D_a f$, proving the displayed range identity. Iteration, boundedness of derivative evaluations, and Hermite interpolation give the last assertion and the exact codimension. $\square$

3. Proof of the classification

Proof of Theorem 1. Factor $q = q_{\mathbb{D}} q_{\mathbb{T}} q_{\text{out}}$ according as its roots lie in $\mathbb{D}$, on $\mathbb{T}$, or outside $\overline{\mathbb{D}}$. Since $\|S\| \leq 1$, every exterior factor is invertible. Lemma 1 says each boundary factor has dense range; the commuting product of these factors also has dense range (equivalently, the product of their injective adjoints is injective). Since $q_{\mathbb{D}} P^2(\mu)$ is closed by Lemma 2, continuity gives

$$[q] = q_{\mathbb{D}} P^2(\mu).$$

The codimension assertion follows from the same lemma.

Now suppose $q \in M \cap \mathbb{C}[z]$ is nonzero. Then $q_{\mathbb{D}} P^2(\mu) = [q] \subset M$, so M has finite codimension. Modulo $q_{\mathbb{D}} P^2(\mu)$, polynomials exhaust the finite-dimensional cyclic module

$$P^2(\mu)/q_{\mathbb{D}} P^2(\mu) \cong \mathbb{C}[z]/(q_{\mathbb{D}}).$$

The image of M is a submodule. Its inverse image in $\mathbb{C}[z]$ is an ideal containing $(q_{\mathbb{D}})$, hence equals (d) for a monic divisor d of $q_{\mathbb{D}}$. Because $d \in M$, invariance and closedness give $dP^2(\mu) \subset M$; the quotient description gives the reverse inclusion. Thus $M = dP^2(\mu)$.

Conversely, if M has finite codimension, let Q be the operator induced by S on $P^2(\mu)/M$. Its minimal polynomial q satisfies $qP^2(\mu) \subset M$, so $q \in M$. The preceding argument applies. (The eigenvalues of Q^* are bounded point evaluations; contractivity places them in $\overline{\mathbb{D}}$ and Lemma 1 excludes $\mathbb{T}$.) Finally, $dP^2(\mu) = [d]$ is cyclic. $\square$

4. The Hardy-equivalent class

Proof of Theorem 2. If K_{ν} is the Carleson embedding constant, then for every polynomial p ,

$$c\|p\|_{H^2}^2 \leq \|p\|_{L^2(\mu)}^2 \leq (C + K_{\nu})\|p\|_{H^2}^2.$$

Thus $P^2(\mu)$ and H^2 are the same topological shift module. Their closed invariant subspaces coincide, and Beurling's theorem writes each nonzero one as $\theta H^2 = [\theta]$.

If μ is rotationally invariant, its nonzero boundary part is c, m . For the radial interior part and $p(z) = \sum a_n z^n$, angular orthogonality gives

$$\int_{\mathbb{D}} |p|^2 d\nu = \sum_n |a_n|^2 \int_{\mathbb{D}} |z|^{2n} d\nu \leq \nu(\mathbb{D}) \|p\|_{H^2}^2.$$

So the hypotheses just proved are automatic. $\square$

5. Scope, novelty, and review focus

Theorem 1 removes all algebraic and finite-codimensional cases, and Theorem 2 handles a natural infinite-dimensional class. The remaining regime is non-Hardy-equivalent, infinite-codimensional, and polynomial-free; index one alone does not bridge this gap.

A bounded exact-title and keyword search on 13 August 2026 found no later solution or this classification. Novelty confidence is modest because the proof uses short module-theoretic arguments.

Human-review focus. Check bounded divided differences in Lemma 2, the passage from dense boundary factors to $[q] = q_{\mathbb{D}} P^2(\mu)$, and the finite cyclic-module ideal argument. Do not promote this as a full answer without treating polynomial-free invariant subspaces.

References

- [1] J. R. Akeroyd, J. B. Conway, and L. Yang, *On Nontangential Limits and Shift Invariant Subspaces*, arXiv:1809.10182 (2018).
- [2] A. Aleman, S. Richter, and C. Sundberg, *Nontangential limits in $P^t(\mu)$ -spaces and the index of invariant subspaces*, Ann. of Math. 169 (2009), 449–490.